Trailer Controller Pinout Map

Hardware Profile: Arduino Mega 2560 | Source: shstrailer::pins.h

1. System Indicators & Fixed Hardware Interfaces

Pin	Hardware Identifier	Direction	Description & Target Peripheral
13	STATUS_LED	Output	Status LED / System Heartbeat Signalling
20	SDA	I/O	Hardware I2C Data Line (DIYables SSD1309 OLED)
21	SCL	I/O	Hardware I2C Clock Line (DIYables SSD1309 OLED)

2. Switch & Push-Button Input Matrix (Debounced)

Pin	Hardware Identifier	Direction	Description & Control Layout
22	L1_SW_A	Input	Light Circuit 1: Toggle Switch Position A
23	L1_SW_B	Input	Light Circuit 1: Toggle Switch Position B
24	L2_SW_A	Input	Light Circuit 2: Toggle Switch Position A
25	L2_SW_B	Input	Light Circuit 2: Toggle Switch Position B
26	L3_SW_A	Input	Light Circuit 3: Toggle Switch Position A
27	L3_SW_B	Input	Light Circuit 3: Toggle Switch Position B
28	L4_SW_A	Input	Light Circuit 4: Toggle Switch Position A
29	L4_SW_B	Input	Light Circuit 4: Toggle Switch Position B
30	LED_STRIP_SW	Input	Dedicated Main LED Strip Control Switch
31	POD_LIGHT_SW	Input	Dedicated Auxiliary Pod Light Control Switch
32	WINCH_UP_SW	Input	Winch Directional Control: UP Button Line
33	WINCH_DN_SW	Input	Winch Directional Control: DOWN Button Line

3. Output Power Channels & Actuators

Pin	Hardware Identifier	Direction	Description & Target Load
34	LIGHT1_OUT	Output	High-Current MOSFET: Lighting Circuit 1 Driver
35	LIGHT2_OUT	Output	High-Current MOSFET: Lighting Circuit 2 Driver
36	LIGHT3_OUT	Output	High-Current MOSFET: Lighting Circuit 3 Driver
37	LIGHT4_OUT	Output	High-Current MOSFET: Lighting Circuit 4 Driver
38	LED_STRIP_OUT	Output	High-Current MOSFET: Main LED Strip Driver
39	POD_LIGHT_OUT	Output	High-Current MOSFET: Auxiliary Pod Light Driver
40	WINCH_UP_OUT	Output	Power Relay Control: Winch Direction UP (Interlocked)
41	WINCH_DN_OUT	Output	Power Relay Control: Winch Direction DOWN (Interlocked)
42	TUBIVATOR_OUT	Output	Auxiliary Power Output: Tubivator Driver Channel